1.35 **a.** Show that the interconnection of the elements satisfies the power check.
b. Find the value of the current through each of the elements using the values of power and voltage and the current directions shown in the figure.

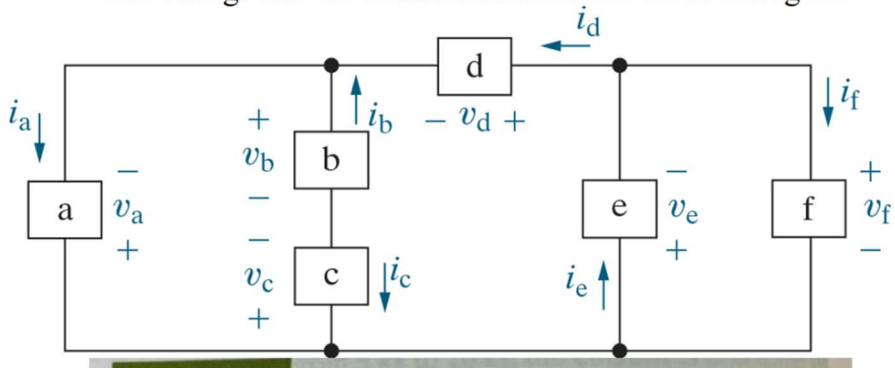


TABLE P1.35

Element	Power (W)	Voltage (kV)
a	750 supplied	−3
b	1600 absorbed	4
c	400 supplied	1
d	150 absorbed	1
e	800 supplied	−4
f	200 absorbed	4

Solution:

P 1.35 [a] We can add the powers supplied together and the powers absorbed together — if the power balances, these power sums should be equal:
$\sum P_{\text{sup}} = 750 + 400 + 800 = 1950$ W;
$\sum P_{\text{abs}} = 1600 + 150 + 200 = 1950$ W.
Thus, the power balances.

[b] The current can be calculated using $i = p/v$ or $i = -p/v$, with proper application of the passive sign convention. Remember that the power supplied is negative and the power absorbed is positive.

$$i_{\text{a}} = -p_{\text{a}}/v_{\text{a}} = -(-750)/(-3000) = -250 \text{ mA};$$

$$i_{\text{b}} = -p_{\text{b}}/v_{\text{b}} = -(1600)/(4000) = -400 \text{ mA};$$

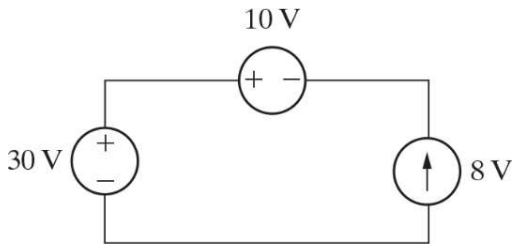
$$i_{\text{c}} = -p_{\text{c}}/v_{\text{c}} = -(-400)/(1000) = 400 \text{ mA};$$

$$i_{\text{d}} = p_{\text{d}}/v_{\text{d}} = (150)/(1000) = 150 \text{ mA};$$

$$i_{\text{e}} = p_{\text{e}}/v_{\text{e}} = (-800)/(-4000) = 200 \text{ mA};$$

$$i_{\text{f}} = p_{\text{f}}/v_{\text{f}} = (200)/(4000) = 50 \text{ mA};$$

2.1 a. Is the interconnection of ideal sources in the circuit in Fig. P2.1 valid? Explain.
b. Identify which sources are developing power and which sources are absorbing power.
c. Verify that the total power developed in the circuit equals the total power absorbed.
d. Repeat (a)–(c), reversing the polarity of the 10 V source.



Solution:

P 2.1 [a] Yes, independent voltage sources can carry the 8 A current required by the connection; independent current source can support any voltage required by the connection, in this case 20 V, positive at the top.

[b] 30 V source: absorbing;

10 V source: delivering;

8 A source: delivering.

[c] $P_{30V} = (30)(8) = 240$ W (abs);

$P_{10V} = -(10)(8) = -80$ W (del);

$P_{8A} = -(20)(8) = -160$ W (del).

$\sum P_{abs} = \sum P_{del} = 240$ W.

[d] The interconnection is valid, but in this circuit the voltage drop across the 8 A current source is 40 V, positive at the top; 30 V source is absorbing, the 10 V source is absorbing, and the 8 A source is delivering.

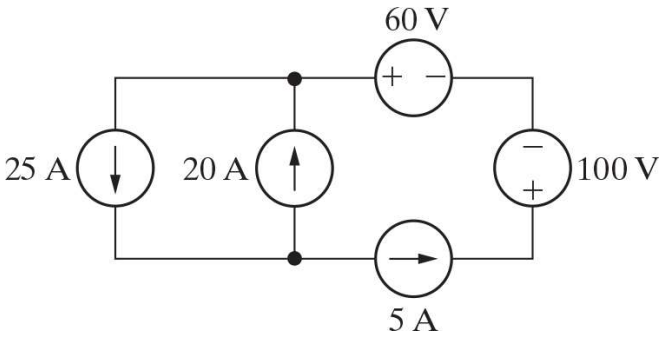
$P_{30V} = (30)(8) = 240$ W (abs);

$P_{10V} = (10)(8) = 80$ W (abs);

$P_{8A} = -(40)(8) = -320$ W (del).

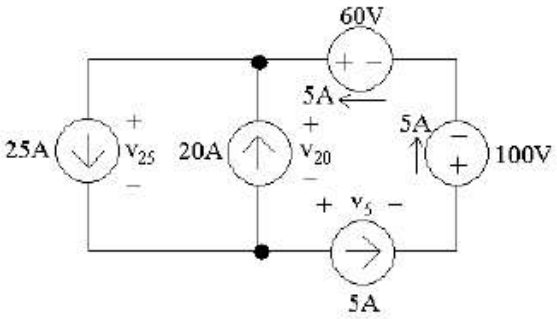
$\sum P_{abs} = \sum P_{del} = 320$ W.

2.3 If the interconnection in Fig. P2.3 is valid, find the total power developed by the voltage sources. If the interconnection is not valid, explain why.



Solution:

P 2.3



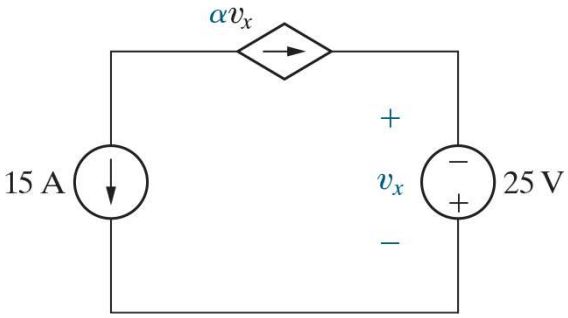
Note that v_5, v_{20}, and v_{25} are not specified. The current sources can carry any amount of current and the voltage sources can carry the 5 A of current. Thus, the interconnection is valid.

We can now calculate the power developed by the two voltage sources:

$$p_{v-\text{sources}} = p_{60} + p_{100} = -(60)(5) + (100)(5) = 200\,\text{W}.$$

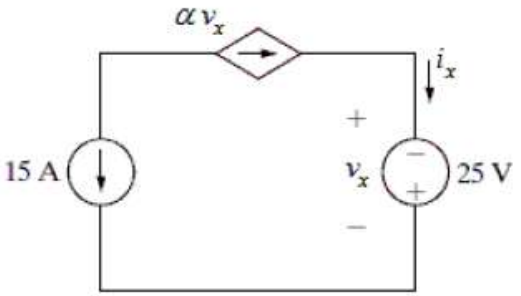
Since the power is positive, the sources are absorbing 200 W of power, or developing −200 W of power.

2.6 **a.** What value of alpha is required in order for the interconnection to be valid?
b. For the value calculated in part (a), find the power associated with the 25 V source.



Solution:

P 2.6



[a] Note from the circuit that $v_x = -25$ V. To find α note that the two current sources are in the same branch of the circuit but their currents flow in opposite directions. Therefore

$$\alpha v_x = -15\,\text{A}.$$

Solve the above equation for α and substitute for v_x,

$$\alpha = \frac{-15\,\text{A}}{v_x} = \frac{-15\,\text{A}}{-25\,\text{V}} = 0.6\ \text{A/V}.$$

[b] To find the power associated with the voltage source we need to know the current, i_x. Note that this current is in the same branch of the circuit as the dependent current source and these two currents have the same direction. Therefore, the current i_x is the same as the current of the dependent source:

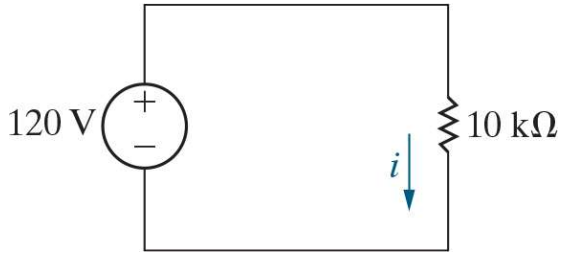
$$i_x = \alpha v_x = (0.6)(-25) = -15\,\text{A}.$$

Using the passive sign convention,

$$p_s = -(i_x)(25\,\text{V}) = -(-15\,\text{A})(25\,\text{V}) = 375\,\text{W}.$$

Thus the voltage source dissipates 375 W.

2.11 **a.** Find i.
b. Find the power supplied by the voltage source.
c. Reverse the polarity of the voltage source and repeat parts (a) and (b).



Solution:

P 2.11 [a] Using the passive sign convention and Ohm's law,

$$i = \frac{v}{R} = \frac{120}{10{,}000} = 0.012 = 12 \text{ mA}.$$

[b] $p_{\text{v.s.}} = p_{\text{R}} = Ri^2 = (10{,}000)(0.012)^2 = 1.44 \text{ W}.$

[c] Using the passive sign convention with the voltage polarity reversed,

$$i = -\frac{v}{R} = -\frac{120}{10{,}000} = -0.012 = -12 \text{ mA};$$

$$p_{\text{v.s.}} = p_{\text{R}} = Ri^2 = (10{,}000)(-0.012)^2 = 1.44 \text{ W}.$$